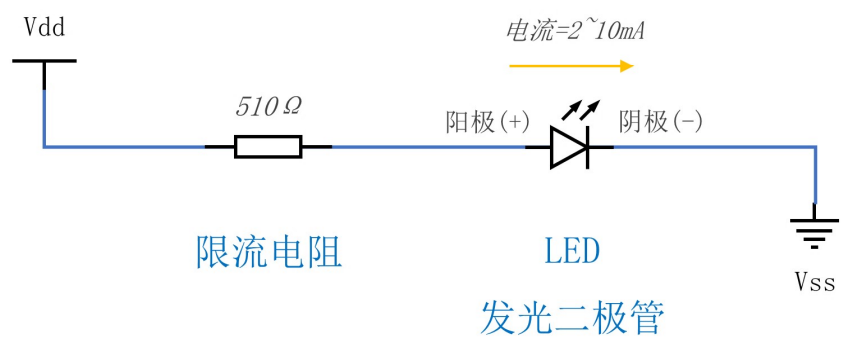


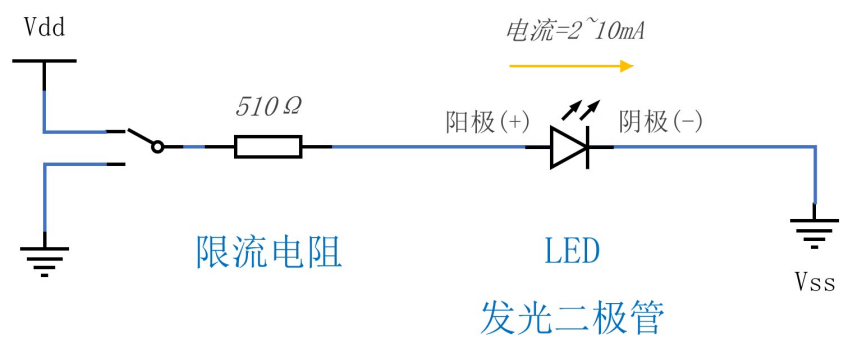


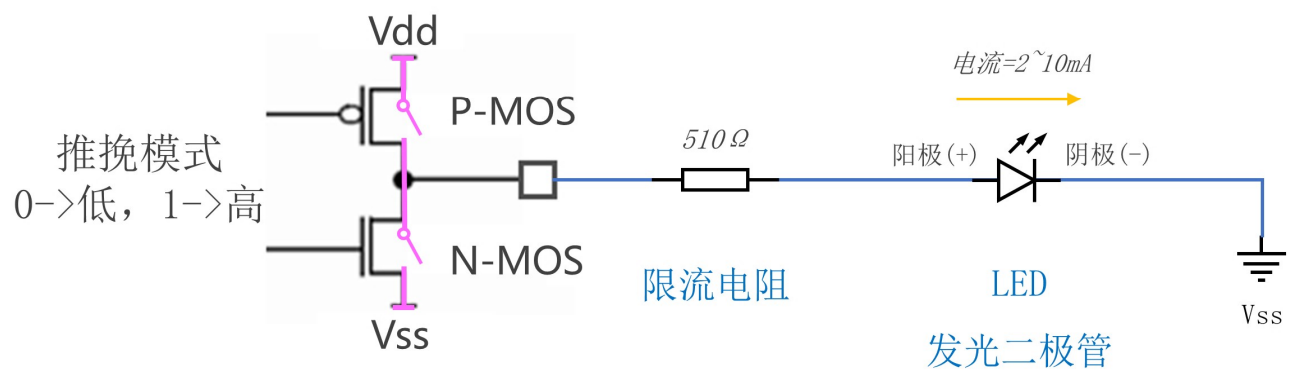
铁头山羊STM32教程

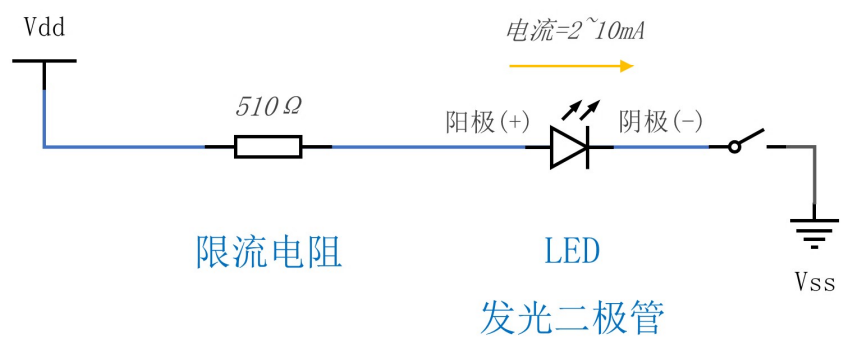
[GPIO]

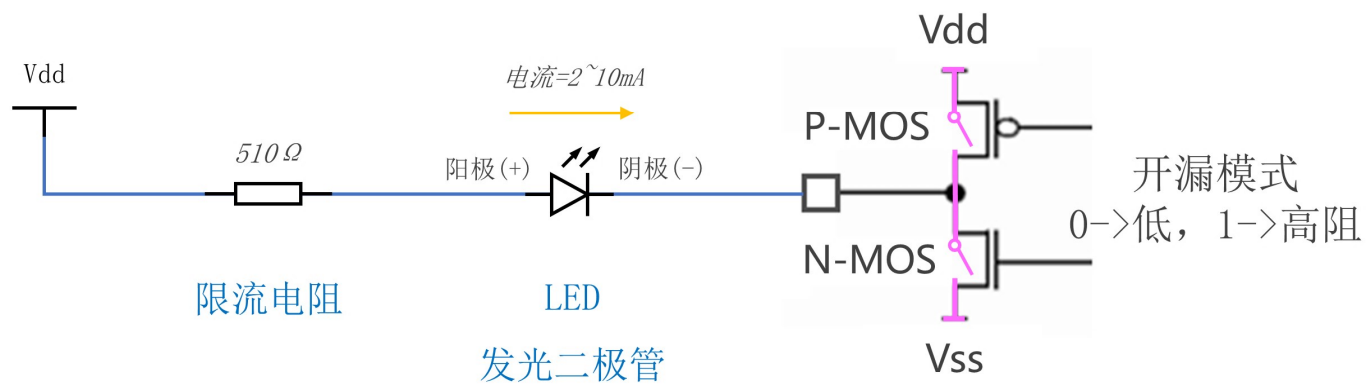
2. 3. LED闪灯实验

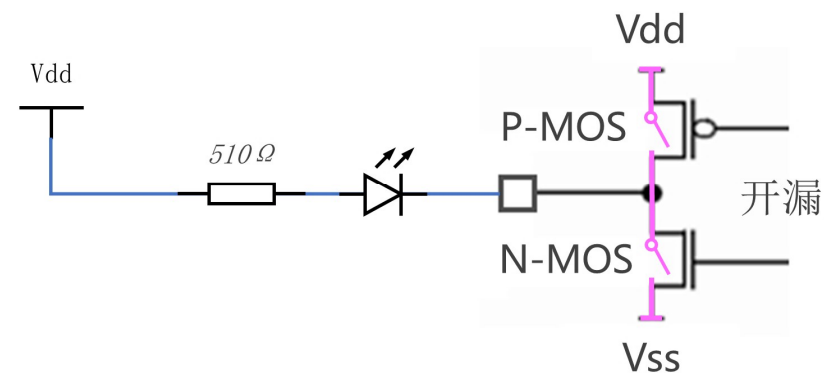
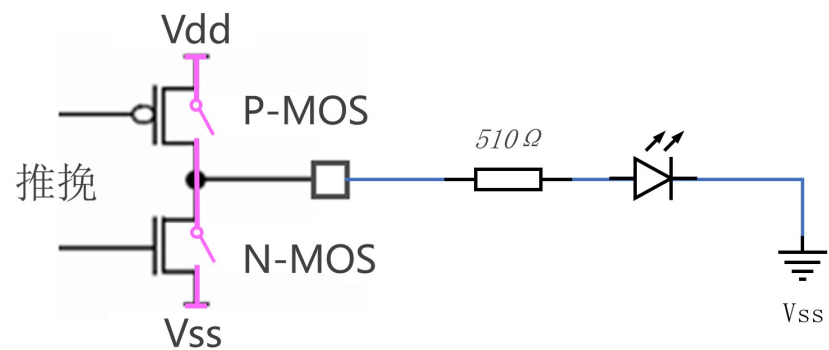
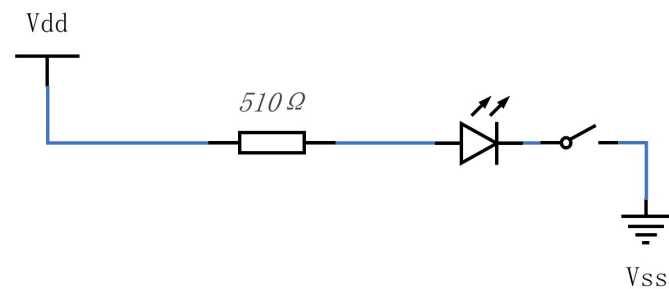
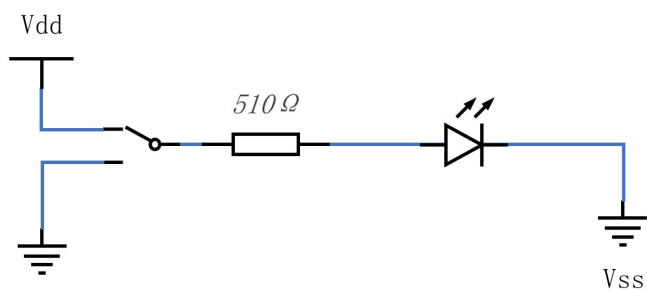


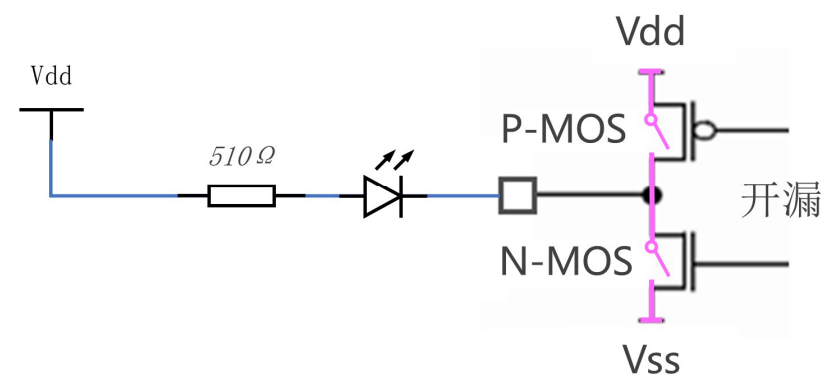
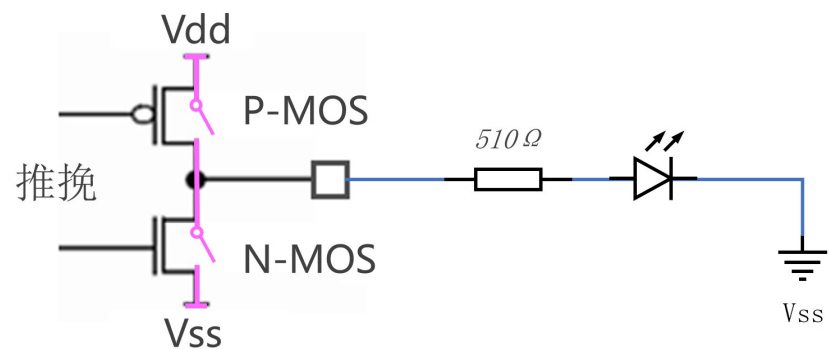
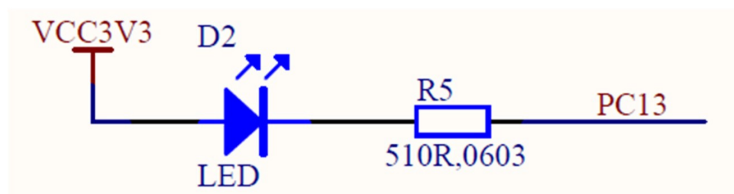














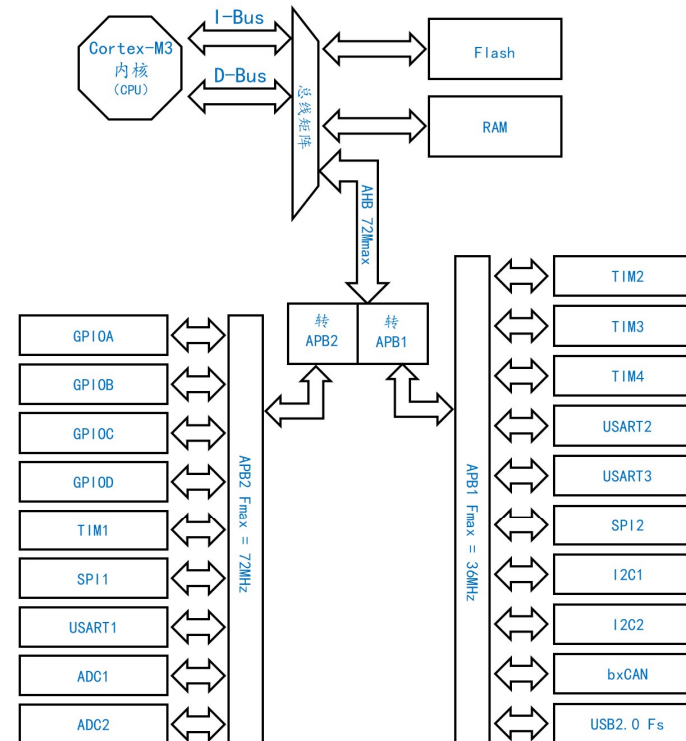
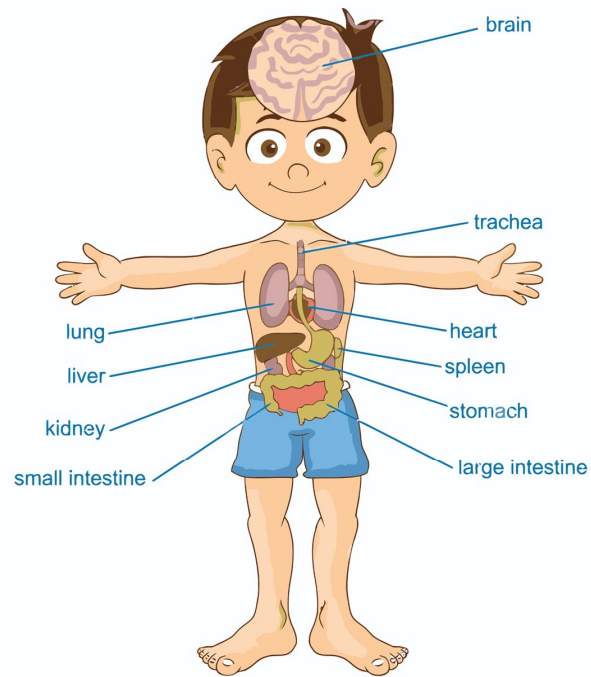
百度网盘地址

<https://pan.baidu.com/s/1kcKd063anDtFZ1gG97c0Kg?pwd=3kvt> 提取码: 3kvt



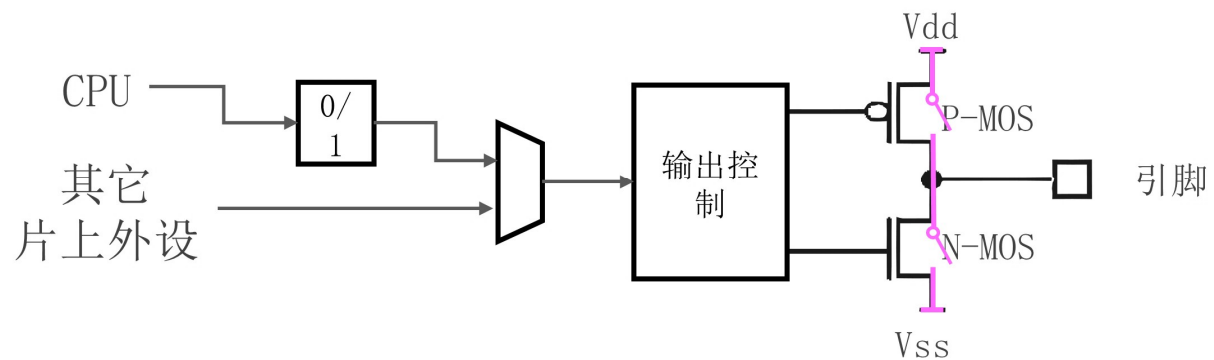
// 开启GPIOx的时钟

```
RCC_APB2PeriphClockCmd(RCC_APB2Periph_GPIOx, ENABLE);
```



GPIO 模块编程接口

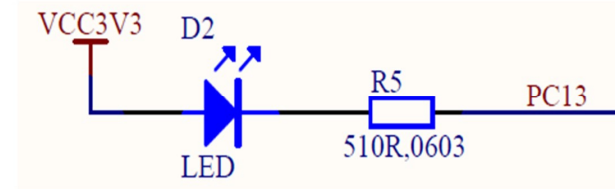
序号	编程接口	作用
1	GPIO_Init	初始化 IO 引脚
2	GPIO_WriteBit	向输出数据寄存器写 1 个比特位
3	GPIO_ReadInputDataBit	从输入数据寄存器读 1 个比特位
4	GPIO_ReadOutputDataBit	从输出数据寄存器读 1 个比特位



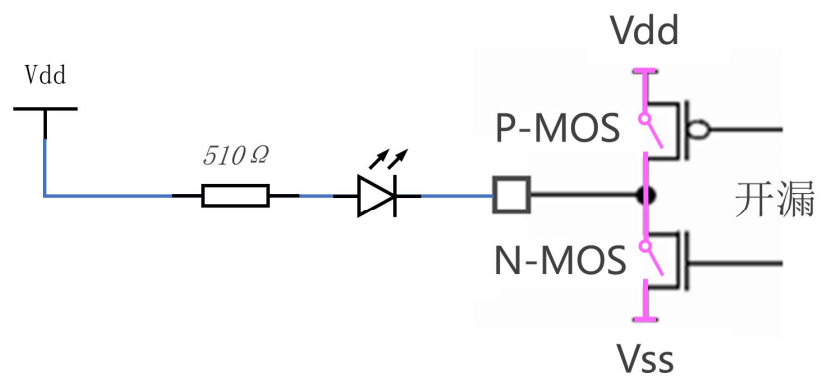
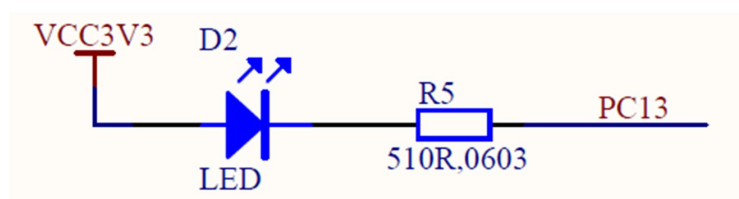
```
void GPIO_Init(GPIOTypeDef *GPIOx, //端口号, 取A B C D..  
                GPIO_InitTypeDef *GPIO_InistStruct); //初始化的参数
```

作用：初始化一枚IO引脚

```
struct GPIO_InitTypeDef{  
    GPIO_Pin; // 引脚编 - GPIO_Pin_0 .. GPIO_Pin_15  
    GPIO_Speed; // 最大输出速度 - GPIO_Speed_2MHz,  
                - GPIO_Speed_10MHz,  
                - GPIO_Speed_50MHz,  
  
    GPIO_Mode; // 模式 - GPIO_Mode_Out_PP 通用输出推挽  
                - GPIO_Mode_Out_OD 通用输出开漏  
                - GPIO_Mode_AF_PP 复用输出推挽  
                - GPIO_Mode_AF_OD 复用输出开漏  
  
}
```

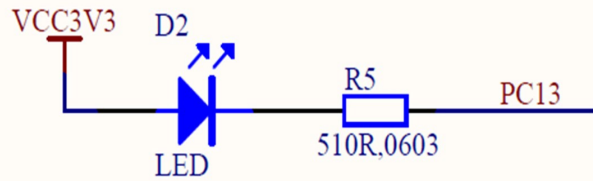


- GPIO_Mode_IPU 输入上拉
- GPIO_Mode_IPD 输入下拉
- GPIO_Mode_IN_FLOATING 输入浮空
- GPIO_Mode_AIN 模拟模式



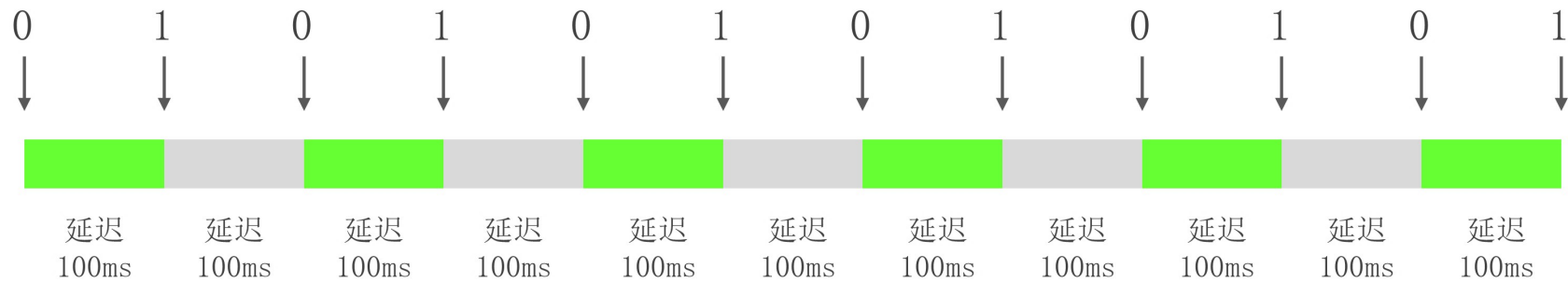
```
void GPIO_WriteBit(GPIOTypeDef *GPIOx, //端口号, 取A B C D..  
                    uint16_t GPIO_Pin, //引脚编号  
                    BitAction BitValue); // 要写入的值  
                                           - Bit_RESET 0  
                                           - Bit_SET   1
```

作用：向IO引脚写值(0/1)



```
GPIO_WriteBit(GPIOC, GPIO_Pin_13, Bit_RESET); // 写0
```

```
GPIO_WriteBit(GPIOC, GPIO_Pin_13, Bit_SET); // 写1
```



```
#include <delay.h>

while(1) {
    GPIO_WriteBit(GPIOC, GPIO_Pin_13, Bit_RESET); // 亮
    Delay(100); // 延迟100ms
    GPIO_WriteBit(GPIOC, GPIO_Pin_13, Bit_SET); // 灭
    Delay(100); // 延迟100ms
}
```